

Gebregziabihier Nigusie Birhane

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Personal Statement

I completed my MSc in IT at Bahir Dar Institute of Technology, with a thesis on NLP for low-resource languages. My research focuses on Machine Learning, Low-resourced NLP, and Computer Vision.

Employment History

- 09/2025 – 06/2026 📌 **AI Engineer for Heritage in Code Project** Wits Innovation Center and MIND Institute, South Africa.
- 08/2022 – 07/2026 📌 **Lecture**, Information Technology Department, Mizan-Tepi University, Ethiopia. Conducting NLP and computer vision research, teaching AI and programming courses, and advising students for their final year project.

Education

- 11/2020 – 08/2022 📌 **MSc** Information Technology, Bahir Dar Institute of Technology, Ethiopia. Thesis: Lexical Complexity Detection and Simplification in Amharic Text Using Machine Learning Approach.
- 10/2015 – 08/2019 📌 **BSc** Information Technology, Mizan-Tepi University, Ethiopia. Project: Android Based Tepi Town Crime Report System.

Research Publications

- 📌 **Nigusie, G., Elbatel, M., & Ayana, Y. (2026).** Contour-Guided Segmentation and X-Ray Image Validation for Pneumonia Detection Using Mobile Deep Learning in Low-Resource Settings. International Workshop on Medical Image Computing in Resource Constrained Settings (MICCAI 2025), 299–309. https://doi.org/10.1007/978-3-032-13654-1_30.
- 📌 **Nigusie, G., & Mossa, N., & Tegegne, T. (2026).** Sentiment Polarity Analysis of Amharic Climate Change Discourse Using Large Language Models. Proceedings of Machine Learning Research, 1–10. <https://proceedings.mlr.press/v302/nigusie26a.html>.
- 📌 **Nigusie, G. (2025).** Supervised Machine Learning based Amharic Text Complexity Classification Using Automatic Annotator Tool. Proceedings of the Sixth Workshop on African Natural Language Processing (AfricaNLP 2025), 4–14. <https://doi.org/10.18653/v1/2025.africanlp-1.2>.
- 📌 **Nigusie, G., & Tegegne, T. (2024).** Applying Hybrid Approach of Complexity Annotator Tool and Machine Learning for Lexical Complexity Detection and Simplification in Amharic Text. International Joint Conference on Artificial Intelligence (IJCAI 2024), 42–49. <https://doi.org/10.24963/ijcai.aai4g.2024/7>.
- 📌 **Nigusie, G., & Tegegne, T. (2023).** Amharic Text Complexity Classification Using Supervised Machine Learning. Artificial Intelligence and Digitalization for Sustainable Development (ICAST 2022), 12–23. https://doi.org/10.1007/978-3-031-28725-1_1.

Research Publications (continued)

- **Nigusie, G., & Tegegne, T. (2022).** Lexical Complexity Detection and Simplification in Amharic Text Using Machine Learning Approach. IEEE 4th International Conference on ICT for Development for Africa (ICT4DA 2022), 1–6. <https://doi.org/10.1109/ICT4DA56482.2022.9971306>.

Conference & Workshop Presentations

- Contour-Guided Segmentation and X-ray Image Validation for Pneumonia Detection using Mobile Deep Learning in Low-Resource Settings, **MICCAI2025**, September 2025, Daejeon, **South Korea**.
- Sentiment Polarity Analysis of Amharic Climate Change Discourse Using Large Language Models, **Deep Learning Indaba 2025**, August 2025, Kigali, **Rwanda**.
- Applying Hybrid Approach of Complexity Annotator Tool and Machine Learning for Lexical Complexity Detection and Simplification in Amharic Text, **Deep Learning Indaba 2024**, September 2024, Dakar, **Senegal**.
- Amharic Text Complexity Classification using Supervised Machine Learning, **10th International conference on Advancement of Science and Technology**, November 2022, Bahir Dar, **Ethiopia**.
- Lexical Complexity Detection and Simplification in Amharic Text Using Machine Learning Approach. IEEE, November 2022, Bahir Dar, **Ethiopia**.

Projects

- **Financial Report Generation** (Mar 2026 – Aug 2026) – Developed an LLM-powered financial reporting pipeline with RAG, vector-based source ranking, SQL generation, and supervised retrieval of financial data and historical records.
- **Heritage in Code** Wits University, South Africa (Nov 2025 – Jun 2026) – Project on AI applications in African music <https://www.wits.ac.za/aiaa/project-teams>
- **Pneumonia Detection using Mobile Deep Learning** September 2025:[App].
- **Mobile-based COVID-19 Information Dissemination** August 2020: [App].

Award & Grants

- AI and African Music Project (\$5,500), Wits University, South Africa, November 2025.
- MICCAI Society MIRASOL Travel Grant (€3,500), Daejeon, South Korea, September 2025.
- Best MSc Thesis Award, Bahir Dar University, Ethiopia, August 2022

Skills

Programming Language	■ Python Javascript Flutter Android Java C++ $\text{\LaTeX}$
ML Tools	■ PyTorch tensorflow HuggingFace
Databases	■ SQL MONGODB MYSQL

References

Available on Request